Attempts and Exploratory Routes

Failed, Partial, and Out-of-Reach Strategies

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This file collects exploratory routes, higher-dimensional extensions, and partial strategies that are currently out of reach or have not led to a closed proof. It is kept separate from the main S^1 sharp-constant argument so that the latter remains focused.

1 Abstract Convolution Side Routes

We first record the parts of the abstract convolution discussion which are useful as orientation, but do not themselves give the desired proof. They explain why the Fredholm route has to use the equation $u = c(u^k * f)$ rather than the formally nearby equation $u^k = c(u * f)$, and why a direct contraction argument is too local to replace the spectral analysis.

1.1 The equation $u^k = c\,(u * f)$: spectral expansion kills solutions

In Fourier space the equation reads $\widehat{u}^{*k}(\xi) = c\,\widehat{u}(\xi)\widehat{f}(\xi)$.

Proposition 1.1 (Type obstruction)**.** *Let $u \in \mathrm{PW}_\tau$ with $\tau > 0$ satisfy $u^k = c(u * f)$ for $k \geq 2$. Then no positive-type solution is possible; every solution has type zero.*

Proof. The Titchmarsh convolution theorem gives $\operatorname{supp}(\widehat{u}^{*k}) \subseteq [-k\tau, k\tau]$, while the right side is supported in $[-\tau, \tau]$. For $k \geq 2$ this forces $\tau = 0$, so u is a polynomial. Degree comparison then forces u to be constant, and the remaining constants are determined by the scalar equation obtained from $u^k = c(u * f)$. □

Corollary 1.2. *For $k \geq 2$ the solution set of $u^k = c(u * f)$ in the Paley–Wiener class is a finite discrete set of constants.*

This is a useful negative result, but it is not a route to the restriction problem. The Euler–Lagrange equation has the smoothing/compactness orientation $u = c(u^k * f)$, not the type-expanding orientation above.

1.2 The small-ball contraction argument

For the compactly oriented equation

$$u = c(u^k * f),$$

one can bypass Fredholm theory in a small ball. If $T(u) := c(u^k * f)$ and u, v lie in a ball of radius R in PW_{τ_f}, Young's inequality gives

$$\|T(u) - T(v)\| \leq |c|\, kR^{k-1} \|f\|_{L^1} \|u - v\|\,.$$

Thus T is a contraction whenever

$$R < (|c|\, k \, \|f\|_{L^1})^{-1/(k-1)}.$$

This gives local uniqueness near sufficiently small solutions, but it does not control the nonlinear eigenfunctions relevant to the sharp restriction constant. In particular, it gives no quantitative isolation around the constant solution at the physical normalisation, and it does not separate the constant eigenvalue from possible nonconstant branches.

1.3 Comparison of the two equations

	$u^k = c(u * f)$	$u = c(u^k * f)$
Fourier identity	$\widehat{u}^{*k} = c\widehat{u}\widehat{f}$	$\widehat{u} = c\widehat{u}^{*k} \cdot \widehat{f}$
Type constraint ($k \geq 2$)	Forces $\tau = 0$ (Titchmarsh)	Allows $\tau \leq \tau_f$
Operator structure	No compactness	T compact
Linearisation	Not Fredholm in general	$I - K$, Fredholm index 0
Solutions ($k \geq 2$, generic)	Finitely many constants	Isolated; Leray–Schauder applies

Table 1: Analytic contrast between the two convolution equations.

The table is useful diagnostically: it shows that the equation with the most rigid support behaviour is the wrong model for the restriction Euler–Lagrange problem, while the correct model has Fredholm structure but requires the harder compactness and spectral-gap inputs.

2 The Higher-Dimensional Transversality Route

For $n \geq 2$ and the equation $\rho = cT(\rho)$ on $L^2(S^{n-1})$, Proposition 3.1 forces $\widehat{u}$ onto S^{n-1}, and the problem reduces to the incidence-fibre equation (3). The symmetry group is $G = \mathbb{R}^n \rtimes SO(n)$, acting on densities by $(x_0, R) \cdot \rho(\omega) = e^{-ix_0 \cdot \omega}\rho(R^{-1}\omega)$.

2.1 The five-step programme

The route to finiteness of shapes requires five analytic inputs. We state each and assess its status for general n.

Step 1: Smoothing / Fredholm estimate. [Open] The target is a multilinear smoothing estimate

$$A_k : H^s(S^{n-1})^k \to H^{s+\varepsilon}(S^{n-1}), \qquad \|A_k(\rho_1, \dots, \rho_k)\|_{H^{s+\varepsilon}} \le C_s \prod_{j=1}^{k} \|\rho_j\|_{H^s},$$

for some $\varepsilon > 0$, where A_k is the k-linear operator behind the incidence-fibre equation. A weaker but sufficient target is compactness of A_k on bounded sets.

For the S^1 problem, the truncation argument (Theorem 3.3) bypasses this step entirely. For $n \ge 2$, the proof must exploit the curvature of S^{n-1} and the transversality of the constraint $\omega_1 + \cdots + \omega_k = \omega$. The key obstruction is the existence of *degenerate fibres*; see Section 2.2.

Step 2: Regularity bootstrap. [Open] Show that weak L^2 solutions are automatically smooth: $\rho \in L^2(S^{n-1}),\ \rho = cT(\rho) \implies \rho \in H^s$ for all s. If Step 1 gives $T : L^2 \to H^\varepsilon$, iteration yields the bootstrap.

Step 3: Local slices for the symmetry action. [Open] At a solution ρ, construct a complement S_ρ to the orbit tangent

$$\mathfrak{g} \cdot \rho = \{-i(a \cdot \omega)\rho - (A\omega) \cdot \nabla_{S^{n-1}}\rho : a \in \mathbb{R}^n,\ A \in \mathfrak{so}(n)\}.$$

This must be done separately on orbit-type strata (radial, zonal, fully asymmetric solutions have different stabiliser dimensions).

Step 4: Nondegeneracy modulo symmetry. [Open] Verify $\ker L_\rho = \mathfrak{g} \cdot \rho$ at each solution, where $L_\rho = I - ckA_k(\rho, \dots, \rho, \cdot)$. For the S^1 problem at the constant solution, this reduces to Bessel integrals (Theorem 4.4).

Step 5: Compactness modulo G. [Open] Prove sequential compactness of the solution set modulo G:

$$\rho_j = cT(\rho_j) \implies \exists g_j \in G :\ g_j \cdot \rho_j \to \rho_\infty.$$

This requires a priori bounds ($\|\rho\| \le R$), a non-collapse lower bound ($\|\rho\| \ge r > 0$), and a no-escape condition for translations. Rotations are compact; translations are the genuine noncompact obstruction.

2.2 Degenerate fibres

The fibre $V_k(\omega) = \{(\omega_1, \dots, \omega_k) \in (S^{n-1})^k : \sum \omega_j = \omega\}$ is generically a smooth manifold of dimension $(n-1)k - (n-1+1) = (n-1)(k-1) - 1$... when the

differential of $\Sigma : \omega_1 + \cdots + \omega_k$ has full rank. Degeneracies occur when all tangent lines $T_{\omega_j} S^{n-1}$ are parallel.

Example for $k = 5$ on S^1: The configuration $(\omega_0, \omega_0, \omega_0, -\omega_0, -\omega_0)$ with output $\omega = \omega_0$ is degenerate. Here $d\Sigma = 3(\text{tangent at } \omega_0) - 2(\text{tangent at } -\omega_0)$ has rank 1 instead of 2 (the tangents at ω_0 and $-\omega_0$ are parallel on S^1). This is a codimension-1 degeneracy, producing an integrable singularity $\sim |t|^{-1/2}$ in the fibre density.

In higher dimensions, the degenerate locus can have higher codimension (tangents at ω_0 and $-\omega_0$ are parallel in $\mathbb{R}^n$ only when $n = 1$; for $n \geq 2$, anti-parallel tangent *spaces* are generically transverse). This suggests that the smoothing estimate may be easier for $n \geq 3$ than for $n = 2$.

2.3 Orbit dimension and radial rigidity

Proposition 2.1 (Orbit dimension, $n = 3$)**.** *For $G = \mathbb{R}^3 \rtimes SO(3)$ ($\dim G = 6$), the orbit dimension of a solution u is $6 - \dim \mathrm{Stab}_G(u)$. Specifically:*

- *Radial: $d = 3$ (only translations).*
- *Zonal (one axis of symmetry): $d = 5$.*
- *Fully asymmetric: $d = 6$.*

Conjecture 2.2 (Radial rigidity)**.** *For the equation $u = c(|u|^{2(k-1)} u * \widehat{\sigma})$ on $\mathbb{R}^3$, every non-trivial solution is rotationally invariant about some point (its "centre of mass"). If true, all solutions belong to 3-dimensional translation orbits, and the shape count reduces to counting radial profiles.*

Warning 2.3. Conjecture 2.2 is currently *unproved*. The preceding sections on the $n = 3$ problem treated it as established; it should be regarded as a useful heuristic until proved.

3 Supplementary: Gegenbauer Expansion and Galerkin Models

3.1 Gegenbauer expansion

[Heuristic]

For completeness, we record the formal harmonic decomposition. Let $\rho(\omega) = \sum_\ell a_\ell C_\ell^\lambda(\omega \cdot \eta)$ with $\lambda = (n-2)/2$. The product of Gegenbauer polynomials linearises as $C_{\ell_1}^\lambda C_{\ell_2}^\lambda = \sum_j \gamma_j(\ell_1, \ell_2) C_j^\lambda$. If T were a spherical convolution, the Funk–Hecke theorem would give $C_\ell^\lambda *_S \sigma = \lambda_\ell(\sigma) C_\ell^\lambda$. Substituting into $\rho = cT(\rho)$ and equating coefficients would give an algebraic system for $\{a_\ell\}$.

However, as noted in Warning 3.3, the incidence-fibre operator is *not* spherical convolution. The Gegenbauer expansion gives a useful *model* but does not rigorously reduce to finitely many coefficients without separate justification.

3.2 Galerkin approximation

Let $H_L = \mathrm{span}\{Y_{\ell m} : 0 \le \ell \le L\}$ and P_L the $L^2(S^{n-1})$ projection. The projected equation $P_L F(\rho_L) = 0$ is a polynomial system of degree k in $N(L) = (L+1)^2$ unknowns (for $n = 3$).

Proposition 3.1 (Galerkin shape count)**.** *After imposing algebraic slice conditions, the number of Galerkin shapes at truncation L is bounded by the BKK mixed volume of the sliced Newton polytopes, in particular by $k^{N(L)}$ (Bézout).*

Warning 3.2 (Limit caveat). This is a statement about the fixed-L model. It does not imply finiteness of true shapes as $L \to \infty$ without uniform a priori bounds, no-escape compactness, and stability of roots under high-mode perturbation.

Remark 3.3 (How Galerkin helps). A Galerkin computation becomes rigorous if paired with a high-mode bound $\|(I - P_L)T(\rho)\|_X \le \varepsilon_L \Phi(\|\rho\|_X)$ with $\varepsilon_L \to 0$, plus an inverse bound on the sliced Jacobian at each computed root. A Newton–Kantorovich argument would then validate nearby true solutions and certify completeness.

4 Supplementary: Validated Search Strategy

Assuming the transversality route is completed (modulo G the solution set is finite), the next goal is to *find every shape* by a finite, checkable search.

4.1 Constants needed

1. **A priori radius:** $\|\rho\|_X \le R(c, k)$.
2. **Non-collapse:** $\|\rho\|_X \ge r(c, k) > 0$.
3. **Regularity/mesh modulus:** $\|\rho\|_{C^m} \le B_m$.
4. **Validated quadrature:** error $q_h(B_m) \to 0$ for the fibre integral on a mesh of size h.
5. **Slice conditioning:** $\|L_\rho \eta\|_X \ge \mu(\rho)\, \|\eta\|_X$ on S_ρ, with $\mu_* := \inf_\rho \mu(\rho) > 0$.
6. **Lipschitz:** $\|DF(\rho) - DF(\tilde{\rho})\| \le M\, \|\rho - \tilde{\rho}\|_X$.

4.2 Separation and packing

From the conditioning bound, the quantitative isolation scale is $\delta_* \simeq \mu_*/M$: two distinct shapes cannot lie within δ_* of each other in the slice. A crude packing bound is

$$\#\{\text{shapes}\} \leq \left(1 + \frac{2R_h}{\delta_* - 2\eta_h - 2q_h}\right)^{N_{\text{mesh}}(h)},$$

provided $\delta_* > 2\eta_h + 2q_h$.

4.3 The search pipeline

1. Choose a quasi-uniform mesh on S^{n-1} (or directly on S^1 for the circle problem) and fix gauge conditions.
2. Represent candidates by interval boxes for nodal values plus global regularity bounds.
3. For each box, enclose the residual $F(\rho)$ at mesh points using validated quadrature.
4. Discard boxes with nonzero residual enclosure.
5. Apply Krawczyk/interval Newton to surviving boxes.
6. Validate uniqueness and lift to the full space.
7. Certify completeness: every box discarded or assigned.

4.4 Main difficulties

1. **Translation gauge:** $e^{-ix\cdot\omega}\rho(\omega)$ creates oscillation as $|x| \to \infty$. A canonical centre must be fixed.
2. **Uniform conditioning:** $\mu_* > 0$ may be the hardest constant to obtain without knowing all solutions.
3. **Mesh consistency:** nodal errors must not hide solutions between mesh points.
4. **Dimension:** the effective search dimension grows with mesh size. Symmetry decomposition and adaptive refinement are essential.

4.5 Specialisation to S^1

For the circle problem, the validated search simplifies considerably:

- The mesh is a grid on $[0, 2\pi)$ with N equally spaced points.

- The gauge fixes phase, rotation of S^1, and one translation.
- The fibre integral reduces to a trigonometric integral.
- The Jacobian involves the linearised operator L, which decomposes into Fourier modes for radial u_0.

The direct certified programme for the sharp constant is:

1. Prove a priori bounds: $\|f\|_{L^2} = 1$, $\|f\|_{C^m} \leq B_m$, $0 < \Lambda_{\min} \leq \Lambda \leq C_{\mathrm{ST}}^6$.
2. Use Lemma 2.3 to restrict the extremiser search to boxes containing a nonnegative representative.
3. Fix gauges (phase, rotation, translation); in the nonnegative slice, Lemma 5.11 fixes the translation parameter.
4. Mesh S^1; represent f by interval boxes plus derivative bounds.
5. For each (f, Λ) box, enclose the residual $E^*(|Ef|^4 Ef)(\omega_i) - \Lambda f(\omega_i)$.
6. Discard or validate via interval Newton.
7. Sort validated Λ_j intervals; certify the ordering.

5 Summary: The Logical Dependency Graph

We summarise the dependencies and status of the full programme.

Proved:

- Spectral support forcing ($\widehat{u} \subseteq S^1$)
- 1D case complete
- Compactness of linearised operator K_{u_0} on $L^2(S^1)$
- Fredholm index zero for $L = K - \Lambda I$
- Eigenvalue decomposition at the constant solution
- Nondegeneracy $\Leftrightarrow$ Bessel integrals (7)
- **Nondegeneracy at the constant: complete proof** (Theorem 4.9): Arb certificate for $2 \leq n \leq 12$, explicit Landau/Krasikov bound for $n \geq 13$
- Spectral gap $\nu_0 \geq 0.4\Lambda_0$ certified
- Regularity bootstrap for critical points
- Inverse Strichartz inequality for S^1 extension
- Profiles of critical sequences are critical points
- Existence of a maximiser (Theorem 5.7)
- Positivity reduction for extremisers (Lemma 2.3)
- Finiteness of nonnegative critical values in compact positive windows (Theorem 6.1)
- NK isolation radius $r_0 \geq 0.13$ (floating; Arb pending)

[Review comment.] The last item should not sit under "Proved" in a final version: it is explicitly floating-point and Arb-pending. A better status split is "proved", "partially certified", and "numerical target". Likewise, the sharp-constant route currently needs a quantitative BTZ stability certificate and a rigorous high-mode interaction bound before the conditional conclusion becomes a proof.

[Response.] **[RESOLVED]** Agreed. The status box will be revised to use three categories: **Proved** (complete proofs), **Partially certified** (correct arguments with floating-point constants pending Arb), and **Numerical target** (computations not yet attempted). The NK radius moves to "Partially certified". The sharp-constant conditional conclusion will list all four pending certificates explicitly: (1) Bessel gap, (2) NK radius, (3) high-mode cross bound, (4) quantitative BTZ stability modulus.

Next concrete steps:

- Certify M_0 and r_0 by Arb ball arithmetic (sixfold Bessel integrals)
- Connect the band-limited uniqueness of Barker–Thiele–ZK (modes ≤ 120) with the bootstrap tail to bypass the far-region branch-and-bound search
- Prove the constant is the unique maximiser modulo G (this alone gives $\mathcal{R}$)

Open (harder):

- Rule out multi-profile splitting for non-maximising critical sequences (Theorem 5.10)
- Enumeration / uniqueness of non-constant shapes

*Notes compiled from working discussions. Statements marked **[Proved]** carry complete proofs; statements marked **[Conditional]** depend on explicitly stated hypotheses; statements marked **[Open]** require new analytic input.*

6 Working notes: failed and partial approaches

This section records approaches that were explored but did not succeed, together with the specific obstructions encountered. The material is kept for reference and to avoid repeating dead ends.

6.1 Approaches to bounding the spectral gap γ_0

The spectral gap $\gamma_0 := \inf\{\Lambda - \Lambda_0 : \Lambda \in \mathcal{V}_+,\, \Lambda > \Lambda_0\}$ measures the distance from the constant's eigenvalue to the next nonneg critical value above it. The finiteness theorem (Theorem 6.1) gives $\gamma_0 > 0$ non-constructively. We attempted several routes to an effective lower bound; none succeeded cleanly.

Cauchy–Schwarz envelope. For nonneg f with $\|f\|_2 = 1$, write $|Ef(r,\phi)| \le 2\pi a_0 |J_0(r)| + 2\pi\sigma_0 S(r)$ where $S(r) := \sqrt{\sum_{n\ge 2} J_n(r)^2}$. By the Neumann addition theorem, $S(r)^2 = (1 - J_0(r)^2 - 2J_1(r)^2)/2$, which is $O(1)$ for large r (no decay). Consequently $\int [a_0|J_0| + \sigma_0 S]^6 r\, dr$ diverges for $\sigma_0 > 0$, and the envelope bound cannot certify $J(f) < \Lambda_0$. **Obstruction:** loss of oscillatory cancellation between Bessel modes.

Leading-order sextic expansion. Grouping the sextuples by number of nonzero modes: $J(f) = \Lambda_0(2\pi a_0^2)^3 + C_2 a_0^4\sigma_0^2 + C_4 a_0^2\sigma_0^4 + C_6\sigma_0^6$, with $C_2 = 15\,\mathcal{J}_2/\mathcal{J}_0 \approx 1.65$ from the $(4 \text{ zero}, 2 \text{ nonzero})$ sextuples. At leading order this gives $\gamma_0 \gtrsim 12$. However, the coefficients C_4, C_6 involve Bessel integrals with 4–6 nonzero indices, whose bounds require the same type of computation as c_{stab} itself. The leading-order estimate is therefore not self-contained.

Truncation of the EL residual. Attempting to certify $T(f) - \Lambda f \ne 0$ mode-by-mode with a Fourier truncation at level N_0: the Lipschitz constant of T is $\sim 10\, C_{\text{ST}}^6 \approx 7000$, while the mode-0 signal $(T(f))_0 \sim \Lambda_0 a_0 \approx 210$. The truncation error exceeds the signal for *every* N_0, because $\sigma_{N_0} \ge c/(2\pi)^{1/2}$ for nonneg f far from f_0. **Obstruction:** the quintic nonlinearity has too large a Lipschitz constant relative to the eigenvalue spacing.

BTZ SOS eigenvalues. The BTZ proof reduces to positive semi-definiteness of matrices Q_D whose entries are sixfold Bessel integrals. The minimum eigenvalue $\lambda_{\min}(Q_0) \sim 0.15\, N^{-1.74}$ (Figure 2 of BTZ-II) certifies $\Lambda_0 - J(f) \geq \lambda_{\min} \sum |v_{\mathbf{m}}|^2$ where $v_{\mathbf{m}} = \widehat{f}_{m_1} \widehat{f}_{m_2} \widehat{f}_{m_3}$ are triple products. Converting to a quadratic stability $c_{\text{stab}}\, d_G^2$ introduces a factor $\widehat{f}_0^4 \sim (2\pi)^{-2}$, giving $c_{\text{stab}} \sim 10^{-6}$—far below the needed threshold. **Obstruction:** the SOS basis (cubic products) is not adapted to quadratic distance from f_0.

6.2 The closure argument and its cost

The approach that *does* work, at least in principle, is the c_{stab} closure from Section 6.6: certify

$$c_{\text{stab}}(N) := \inf_{\substack{g \text{ real, nonneg, } \|g\|_2=1 \\ \widehat{g}(n)=0 \text{ for } |n|>N,\, g \neq f_0 \bmod G}} \frac{\Lambda_0 - J(g)}{d_G(g, f_0)^2} \;>\; \text{threshold}(N),$$

where $\text{threshold}(N) := (C_Q - 3)\Lambda_0 \sigma_N^2 / (r_0 - \sigma_N)^2$ with $\sigma_N = A_3/(\sqrt{\Lambda_-}\, N^{5/2})$ from Corollary 6.9.

Both sides depend on N. The threshold decays as N^{-5} (from σ_N^2). Floating-point scans give $c_{\text{stab}}(2) \geq 128$ and $c_{\text{stab}}(4) \geq 39$, suggesting a decay $c_{\text{stab}}(N) \sim C_0 N^{-\alpha}$ with $\alpha \approx 1.7$ (fitted from two points; consistent with the BTZ eigenvalue decay $\sim N^{-1.74}$).

Since $\alpha < 5$, the threshold eventually drops below c_{stab} for N large enough. The crossing point N_{cross} and the associated BTZ computational cost $\sim N_{\text{cross}}^5$ depend on the actual value of α:

α	N_{cross}	BTZ cost ($\sim N^5$)
0 (no decay)	~ 140	$\sim 5 \times 10^{10}$
1	~ 330	$\sim 4 \times 10^{12}$
1.74 (empirical)	~ 900	$\sim 6 \times 10^{14}$
2	~ 1400	$\sim 6 \times 10^{15}$
3	~ 27000	$\sim 10^{22}$

For comparison, the published BTZ computation at $N = 120$ used 94 compute nodes for 16 hours with $\sim 1.6 \times 10^8$ Bessel integrals.

Next steps. The decay rate α can be pinned down by computing $c_{\text{stab}}(N)$ at $N = 6, 8, 10, \ldots$ via the global stability scan (`numerics/global_stability_v2.py`). Once α is known to one decimal place, the required N is determined, and the computational cost can be quoted to a numerics team for a BTZ-type Arb computation at level N_{cross}.